

SYNOPSIS REPORT

Parkinson's Disease Progression Prediction Using Gait Analysis and Wearable Sensor Data

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Computer Science and Engineering

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1. INTRODUCTION

Parkinson's Disease is a progressive neurological disorder that primarily affects human movement. It develops gradually and worsens over time due to the degeneration of nerve cells responsible for dopamine production. As a result, patients experience tremors, stiffness, slow movement, and balance issues.

Among all symptoms, gait abnormalities play a crucial role in identifying disease progression. Changes in walking patterns such as reduced stride length, slower walking speed, and irregular step patterns are strong indicators of Parkinson's severity.

Traditional monitoring methods rely heavily on periodic clinical observations. These methods are not continuous and often fail to capture real-world patient movement. With the advancement of wearable sensors, it is now possible to continuously collect gait data in real-time.

This project focuses on analyzing such gait data using machine learning techniques to predict disease progression stages accurately and efficiently.

2. PROBLEM STATEMENT

Existing Parkinson's disease monitoring systems are limited in providing continuous and objective assessment of patient movement. Most clinical approaches depend on subjective judgment and periodic evaluation, which may not reflect the true condition of the patient.

Therefore, there is a need for a system that can continuously analyze gait patterns using wearable sensor data and automatically predict the progression stage of Parkinson's disease with high accuracy.

3. LITERATURE REVIEW

Various studies have explored the use of wearable sensors and machine learning for Parkinson's disease detection. Researchers have used accelerometer and gyroscope data to analyze movement patterns and identify abnormalities.

Traditional machine learning models such as Logistic Regression, Decision Trees, and Support Vector Machines have been widely used. These models perform well when features are properly engineered.

Recent studies have also explored deep learning approaches such as CNN and LSTM, which are capable of capturing temporal dependencies in gait data. However, these models require large datasets and significant computational power.

Despite these advancements, many existing works lack a combination of strong feature engineering, clustering-based stage discovery, and comparison of multiple models within a unified framework.

This project attempts to bridge this gap.

4. OBJECTIVES

- To preprocess raw gait data from wearable sensors
 - To convert raw sensor data into structured format (CSV)
 - To extract meaningful gait features
 - To discover disease stages using clustering
 - To train multiple machine learning models
 - To implement deep learning model for comparison
 - To visualize results for better interpretation
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5. METHODOLOGY

The project follows a structured pipeline consisting of multiple stages.

5.1 Data Conversion (TXT to CSV)

The original dataset is in raw text format. A preprocessing script is used to convert this unstructured data into CSV format. This step ensures compatibility with data analysis tools.

5.2 Data Cleaning

The dataset is cleaned by removing noise, handling missing values, and ensuring consistency. This step improves the reliability of further analysis.

5.3 Feature Engineering

The cleaned data is divided into segments, and multiple features are extracted:

- Mean and Standard Deviation
- Symmetry and Balance
- Range of movement
- Total Pressure and Force Variation
- Walking Speed
- Stride Length
- Step Variability

These features represent both statistical and biomechanical properties of gait.

5.4 Stage Discovery

K-Means clustering is applied to group the data into different stages of Parkinson's disease. This step automatically identifies patterns without predefined labels.

5.5 Model Training

Multiple machine learning models are trained:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine

Additionally, a CNN-LSTM model is used to capture sequential patterns in data.

5.6 Data Visualization

Visualization plays a key role in understanding model performance. The project includes:

- Confusion matrix
- Feature importance graph
- PCA-based stage visualization

These visualizations help in interpreting results clearly.

6. TOOLS & TECHNOLOGIES

The project is implemented using the following tools:

- Python: Main programming language for development
- Pandas: Used for data manipulation and preprocessing
- NumPy: Used for numerical computations
- Scikit-learn: Used for machine learning models
- TensorFlow/Keras: Used for deep learning model implementation
- Matplotlib: Used for data visualization
- VS Code: Development environment for coding and debugging

These technologies provide a complete ecosystem for building, testing, and evaluating the system.

7. SCOPE

The scope of this project includes:

- Gait data analysis using wearable sensor data
- Prediction of Parkinson's disease progression stages
- Comparison of multiple machine learning models

Limitations:

- Does not include real-time wearable integration
 - Limited dataset size
 - Deep learning performance may vary due to data constraints
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8. USE OF ENGINEERING

This project applies core engineering concepts in a practical manner:

- Data preprocessing and signal cleaning techniques
- Feature extraction from raw sensor signals
- Implementation of machine learning algorithms
- Development of deep learning architectures
- Visualization and performance evaluation

These steps demonstrate how theoretical knowledge is applied to solve real-world healthcare problems.

9. EXPECTED OUTCOMES

- Accurate classification of disease stages
 - Identification of best-performing model (Random Forest)
 - Improved understanding of gait-based diagnosis
 - A complete pipeline for disease prediction
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10. APPLICATIONS

- Early detection of Parkinson's disease
 - Remote patient monitoring
 - Clinical decision support systems
 - Healthcare analytics
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11. CONCLUSION AND FUTURE WORK

The project successfully demonstrates how gait analysis combined with machine learning can be used to predict Parkinson's disease progression.

Future improvements may include:

- Integration with real-time wearable devices
 - Use of larger datasets
 - Development of mobile-based applications
 - Enhancement of deep learning models
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Signature of Supervisor: ____

Date: _____